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Spiral Eddies

In 1872, the British vessel HMS Challenger embarked on a three-year scientific expedition, circumnavigating the globe. This event marked the birth of modern oceanography. In the present age of Earth exploration from space, NASA has acknowledged this early effort by naming one of its Space Shuttles after this pioneering research vessel. In October 1984, the modern Challenger explored the oceans from the vantage point of space. Included in the crew was Paul Scully-Power, a civilian oceanographer at the Naval Underwater Systems Center. (See: Strait of Gibraltar.)

Scully-Power took this photograph from an altitude of 45 km (28 mi) over the central Mediterranean Sea; it covers an area 210 km (130 mi) on a side. Sunlight reflected toward the camera yields a surprising sunglint pattern. The brightness of the sunglint is related to the texture of the ocean surface. In the central portion of the photograph strongly illuminated by the sun, smooth surfaces reflect light as a mirror and appear bright, while regions roughened by short surface waves scatter the sunlight and appear darker. Away from the strongly-illuminated portion, the pattern is reversed with smooth regions appearing darker and rougher regions appearing lighter.

In several places, bright bands wrap up into spiral eddies (1, 2, 3). (These spiral eddies are similar to patterns of shear-flow turbulence created in laboratory experiments.) The wakes of several ships can be seen as horizontal lines crossing the banded structures (4). Note that the wakes exhibit marked displacements as they cross certain bands (5), indicating the presence of strong current shears. The amount of offset of the ships' wakes is dependent on the strength of the current shear and the time elapsed since the ship traversed the area.

In previous Shuttle flights, occasional examples of such spirals had been photographed, but they were believed to be relatively isolated. Perhaps the most notable aspect of observations made from the Challenger is that the spiral eddies and their associated shear currents are quite widespread, occurring throughout much of the open ocean and confined seas. Moreover, their rotation is primarily cyclonic (counterclockwise in the Northern and clockwise in the Southern Hemisphere), and their diameter is consistently on the order of 12 to 15 km (seven to nine miles).

Considerable research is needed before scientists can understand the dynamics of these spiral eddies and their oceanographic significance. Now that they are known to be relatively widespread, experiments using research aircraft and ships can be designed to answer questions such as: How are they generated and maintained? How deep do they extend into the water? How do they interact with other scales of motion? This picture is an example of how observations from the Space Shuttle can be used to suggest potentially significant avenues for follow-on oceanographic research.

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